IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 6 May 2025, accepted 23 May 2025, date of publication 29 May 2025, date of current version 12 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3574788

Daily Crude Oil Prices Forecasting Using a Novel Hybrid Time Series Technique

HASNAIN IFTIKHAR[1,2,6], MOIZ QURESHI[2,3], PAULO CANAS RODRIGUES[4], MUHAMMAD USMAN IFTIKHAR[5], JAVIER LINKOLK LÓPEZ-GONZALES[6], (Member, IEEE), AND HASNAIN IFTIKHAR[5]

[1]Department of Statistics, University of Peshawar, Peshawar 25120, Pakistan
[2]Department of Statistics, Quaid-i-Azam University, Islamabad 45320, Pakistan
[3]Department of Statistics, University of Sindh, Jamshoro, Sindh 76080, Pakistan
[4]Department of Statistics, Federal University of Bahia, Salvador 40170-110, Brazil
[5]Faculty of Science, Engineering and Built Environment, Deakin University, Burwood, VIC 3125, Australia
[6]Escuela de Posgrado, Universidad Peruana Unión, Lima 15468, Peru

Corresponding authors: Paulo Canas Rodrigues (paulo.canas@ufba.br) and Hasnain Iftikhar (hasnain@stat.qau.edu.pk)

ABSTRACT This paper introduces a new hybrid time series forecasting technique to obtain an efficient and accurate daily crude oil prices forecast. The proposed hybrid technique combines the features of various regression, time series, and machine learning models to improve forecast accuracy. First, it involved processing the original crude oil prices time series to address missing values, variance stabilization, and normalization. Second, it divides the purified crude oil price time series into two major parts: the nonlinear long-term trend (long-run fluctuations) and the residual (short-term fluctuations). The secular non-linear trend series is modeled using various regression models, including linear, spline, lowess, and smoothing spline regression models. On the other hand, the short-run random part is modeled and forecasted using four benchmark time series (linear and non-linear AR, ARMA, ESM) and four machine learning (Neural network autoregressive, Random forest, support vector regression with polynomial and radial basal functions) models. The forecast from both parts is summed to obtain the final forecasting results. The proposed hybrid time series method builds a strong emphasis on effectively measuring the nonlinear long-term trend, which has been disregarded in prior works. A variety of evaluation metrics are employed to evaluate the proposed effectiveness. The testing findings demonstrate the hybrid forecasting technique's accuracy and efficacy. Specifically, an MAE of 1.28106, an RMSE of 1.59259, a PCC of 0.94158, and a DS of 0.82149 are obtained for the Brent and WTI oil markets when the Lowess regression and NPAR models are combined. With regard to WTI oil prices in particular, the model's resilience in producing precise forecasts is further demonstrated by the lowest MAE of 1.25730, RMSE of 1.55390, and PCC of 0.93890.

INDEX TERMS Crude oil price forecasting, regression models, time series models, machine learning models, hybrid time series forecasting technique, decision making.

I. INTRODUCTION

The world's oil market benchmarks are West Texas Intermediate (WTI) and Brent crude oils, which are considered significant trading classes of sweet, light crude oils. Brent and WTI crude oil markets have comparatively low density, which makes them appear light. The low sulfur level of this crude oil gives it a sweeter taste than sour crude oils, and it is also easier to refine. Additionally, both oils are considered two of the most significant variables that have a massive impact on global financial markets, especially on the price of crude oil. Because oil prices affect several markets, such as bond markets, stock markets, etc., investors and academicians studying the oil market have always paid close attention to producing a reliable estimate of the oil price and its volatility [1], [2], [3], [4]. Reliable oil price forecasting interests central banks and policymakers in several nations since it enables them to prepare adequate responses for

The associate editor coordinating the review of this manuscript and approving it for publication was Bang L. H. Nguyen.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

potential shocks resulting from swings in oil prices [5], [6], [7], [8]. Therefore, it would appear essential to make an accurate oil price projection.

Numerous theoretical frameworks exist for predicting and forecasting oil prices. The relevant literature has two essential classes of forecasting models. Classical econometric and statistical methods such as random walk (RW), (ARIMA), (ESM), linear-regression (LinR), (GARCH), and error correction models (ECM) are included in the first class. Although these approaches fall somewhat short of capturing the nonlinear aspects of crude oil, they help capture the linear time series data. Artificial intelligence (AI) models are thus recommended as a solution to this limitation [9], [10], [11], [12], [13].

Historical data and time series models are used to carry out the forecasting process, and this is accomplished by utilizing both linear and non-linear time series models. It is sometimes asserted that it can be challenging to discern between a linear and nonlinear correlation, suggesting that identifying a correlation's linearity or nonlinearity can be a complicated procedure [5], [14], [15], [16]. However, no single forecasting model can produce excellent results under all circumstances because each real-world problem has a unique set of complexities. Not every relationship (correlation) in a time series can be captured by a single model [17], [18], [19], [20]. Therefore, rather than employing a single model, use a compound of many appropriate models [21], [22], [23], [24]. It may be possible to improve the accuracy of the final forecasting using a hybrid model. Additionally, it is easier to capture all kinds of correlations in time series when a combination of forecasting models is used [25], [26], [27], [28].

Many studies on predicting oil prices have been released in recent years. The international oil market's small-scale forecasting model was initially proposed by [29]. Used a non-linear regression approach [30] to estimate the OPEC basket price. An ARIMA and GARCH model for daily oil futures price forecasting was proposed by [31]. Reference [32] used an ARIMA model in this manner to anticipate the price of crude oil, and they then contrasted the outcomes with those of SVM and ANN. Furthermore, a study comparing back-propagation(BP) and SVM analytically for forecasting crude oil prices was led by [33]. In this work [34], a novel methodology is proposed that demonstrates a range of variations in WTI crude oil data. According to findings, the price of WTI crude oil would have a temporary shock upward inclination, but the increase is unlikely to be considerable. To forecast the prices of WTI, Brent, and OPEC crude oils, this investigation [35] uses a semi-parametric Markov-switching AR-ARCH model. As a result of their superior forecasting accuracy over both in-sample and out-of-sample horizons, the semi-parametric Markov-switching models outperform the ARIMA and GARCH models in the estimate results. A self-optimizing ensemble learning model called ICEEMDAN-SCA-RVFL is developed to predict crude oil prices in this work [36]. It incorporates the (ICEEMDAN), (SCA), and (RVFL) neural networks. According to the results, the suggested ICEEMDAN-SCA-RVFL considerably outperforms the individual and group benchmark models when using the historical series of the daily spot prices for crude oil (WTI).

The work in [37] used the Google Index (GI) as an external factor in the ARIMA and ARMA-GARCH models to examine the impact and prediction ability of the index on crude oil prices. According to the empirical findings, the Google Index hurts crude oil prices; however, it is not a reliable indicator of future prices for crude oil. In this work [38] utilized data from over 40,500 observations; estimated and quantified the volatility of high-frequency (WTI) oil prices over a month. This work proposed a novel stochastic volatility model using the non-linear Kalman filter to extract latent volatility.

To forecast crude oil prices, [39] suggested using an ensemble learning framework based on least squares SVM with undetermined parameters. To accurately forecast Brent oil prices, [40] introduced a forecasting model utilizing the wavelet transform and least squares SVM. Likewise, [41] also suggested a wavelet decomposed ensemble (WDE) model to enhance the accuracy of forecasting crude oil prices. In this way, a technique for predicting crude oil prices called SDAE-B, based on deep learning and ensemble learning, was presented by [1]. Additionally, The monthly costs of Henry natural gas (HNG) and heating oil are also examined by [42], along with the daily prices of WTI and Brent crude oil using a non-linear ANN model. Authors in [43] to accurately capture different aspects of fluctuations in crude oil price forecasting. The findings indicate that the blending-based model that has been suggested performs better than the current models in terms of prediction errors for forecasting horizons that are both short and medium. Using an agnostic (RF) model based on 1000 regression trees, [44] analyzed the changes in oil prices. This work demonstrated a 65% reduction in in-sample (RMSEs) using this flexible yet highly disciplined computational approach.

In order to identify the existence of linear and nonlinear correlations between particular input variables and the price of crude oil, [45] employed a combination of a generalized additive (GAM) model with a linear transfer function and ARIMA residuals. The findings show that the most crucial statistic for understanding the movement of oil prices is the physical market balance, or fundamental. A combined forecasting approach, introduced by [46], combines improved (SVR) with a quadratic decomposition. Based on empirical data, the proposed model presented increases in the accuracy of the crude oil price and outperforms other models. Authors in [47] are to determine how particular temporal patterns can affect the dynamics of the WTI and Brent crude oil markets by examining seasonal variations in crude oil returns. On the other hand, the work [48] proposes a unique combined forecasting approach with time-varying weights,

namely the Jaynes weight hybrid (JWH) model, which incorporates the Shannon information entropy (SIE) and numerous forecasting methods, to improve the prediction accuracy and resilience of forecasting models. The outcomes demonstrated that, in comparison to previous models, the suggested model has produced satisfactory results. However, The objective of [49] was to find novel mixed genetic algorithm approaches for oil price forecasting. The seven stages of the Sandelowski and Barroso technique were followed in addition to a systematic and meta-composite review methodology to identify practical elements. The purpose of this work [43] is to foresee crude oil prices by combining heterogeneous predictors. Regression trees, ridge regression, SVR, KNN-R, linear regression, and other machine learning techniques are utilized to create a forecasting model through blended ensemble learning. In the research study [50], a novel hybrid technique, namely mix-kernel extreme learning machine (Mix-KELM) and median ensemble empirical mode decomposition (MEEMD), is proposed to forecast and anticipate the price of crude oil. According to the findings, the suggested approach has the lowest MAE, SMAPE, and RMSE. However, predictive performance differs across markets, and each model often has its own unique structural and functional shape.

This work is a novel hybrid time series crude oil price forecasting; the proposed techniques are based on regression, Time series (TS), and machine learning (ML) models. However, in the proposed forecasting time series technique, the main focus is examining the long-term secular trend component, which is often neglected. Initial analyses suggest that we can significantly improve daily forecasting results by separately modeling and predicting long-term and short-term fluctuations in crude oil prices. Thus, the key contribution of this paper is an in-depth investigation into the impact of accurately estimating the long-term secular trend on daily out-of-sample crude oil price forecasting. We compare the models' forecasting ability within our proposed forecasting framework, considering both classic time series and machine learning, as well as the regression models. Additionally, we conduct a detailed analysis of the variations in prediction accuracy among the considered models.

Following is the framework of the remaining portion of the article: The general process of the suggested hybrid forecasting approach is described in Section II. A practical implementation of this modeling methodology is demonstrated in Section III. A quick comparison between the suggested superior models and other baseline forecasting models is provided in Section IV. Finally, some closing thoughts and recommendations for future study directions are included in Section V.

II. THE PROPOSED HYBRID TIME SERIES FORECASTING TECHNIQUE

To provide daily forecasts of crude oil prices, this section describes the proposed TS framework and projecting method, emphasizing both Brent and WTI prices. The crude oil price time series generally exhibits missing data points, high volatility, non-constant mean and variance, outliers, a regular non-linear trend component (long-run fluctuations), and a random component (short-term fluctuations). To achieve this, First, it involved processing the original crude oil prices time series to address missing values, outliers (extreme values), variance stabilization, and normalization. Second, to get the same pattern time series, There are two main components to the price time series for refined crude oil: the nonlinear long-term trend (long-run fluctuations) and the residual (short-term fluctuations). The nonlinear secular trend series is modeled using various regression models, including linear, spline, lowess, and smoothing spline regression models. However, the short-run random part is modeled and forecasted using four TS (linear and non-linear AR, ARMA, ESM) and four ML (NNAR, RF, SVR with polynomial and radial basal functions) models. The forecast from both parts is summed to obtain the final forecasting results. Descriptions regarding this framework are given in subsections.

A. SETTING UP PRIMARY DATA SET

The main goal of this work is to produce a reliable and effective daily price forecast for Brent and WTI, the two main crude oil markets. To address this objective, we implemented a novel time series forecasting methodology to model and estimate the crude oil prices time series. The author's goal is to comprehend the complex mechanisms that underlie changes in the price of crude oil over time. These features are supposed to comprise missing data points, extreme values (outliers), non-constant mean and variance, a long-run trend, short-run fluctuations, nonstationarity, and nonnormality. For example, these characteristics can be seen in Figures 1-2. Figure 1 (a) for the daily Brent crude oil, and Figure 1 (b) WTI crude oil time series from 2001-January-01 to 2023-August-31. It can show a nonlinear secular trend component in both crude oil time series. Figures 1 (c) and (d) show the box plots and histograms with and without extreme values. It can be observed that the outliers have existed in both crude oil prices time series (Brent and WTI). On the other hand, Figure 2 (1st column) for the ACF plot of the original crude oil prices time series (Brent and WTI) at thirty lags, and Figure 2 (2nd column) for the partial autocorrelation plot of both crude oil prices time series (Brent and WTI) at thirty lags. These visual representations clearly illustrate a discernible nonlinear secular long-run trend component, nonnormality, extreme values, and nonstationarity, which are also evident from these figures.

The goal of this work is to produce extremely precise and effective one-day-ahead crude oil cost projections by addressing these inconsistencies before modeling and forecasting. First, a simple linear interpolation technique replaces some missing data points with the arithmetic mean of the three surrounding observations. When examining crude oil prices, it is imperative to recognize and manage anomalies, such as spikes or surges. In the present investigation, anomalies in the first crude oil price sequence are identified by employing

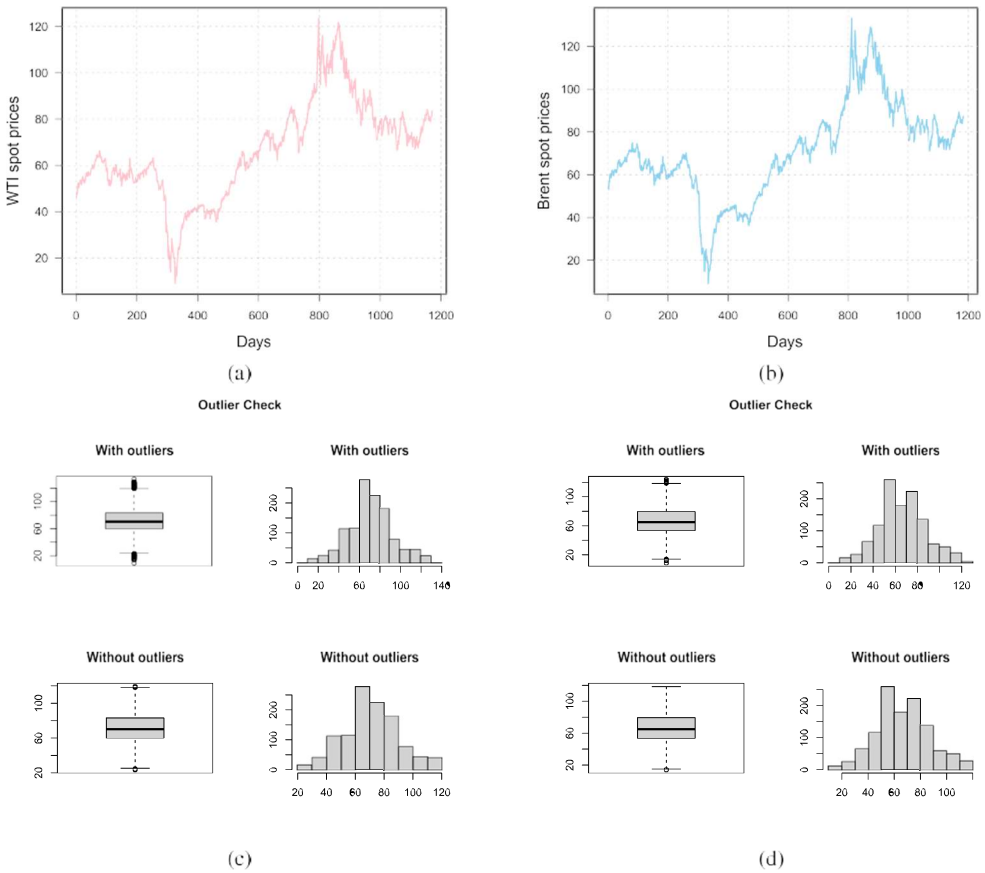

FIGURE 1. Daily histogram of Crude Oil Markets: Brent Oil (a) and WTI Oil (b); Visual outliers (with and without) description for Brent Oil prices (c) and the WTI Oil prices (d).

the 99.7% confidence interval ($\mu \pm 3s$), in which numbers beyond this range are regarded as not normal (outliers). This interval, $\mu \pm 3s$, is defined by the mean of the crude oil price time series, μ, and the standard deviation, s. The method used in practice is to check each data point in turn, and an outlier is a price that is found to be outside of the confidence interval. Next, the closest range value takes the position of this outlier. Notably, replacing outliers lowers the standard deviation; hence, the standard deviation needs to be recalculated every time an outlier is removed. The series is filtered as a result of this operation. Ultimately, the s^2 or s are stabilized by using the logarithmic transformation.

After obtaining the filtering of the crude oil prices in the first three steps, the purified oil price TS (log (O_i^j)) is divided into two major sub-components. The fourth step is the long-term fluctuations (secular trend sub-component; (l_i^j)) and short-term fluctuations (a residual sub-component; (s_i^j)). Therefore, the dynamics of the daily oil prices time series is mathematically defined as:

$$\log(O_i^j) = l_i^j + s_i^j \tag{1}$$

In the above-proposed equations, $\log(O_i^j)$ show the time series of the crude oil prices, $i = 1, 2, 3, \ldots, n$ days, while superscript $j = 1, 2$ indicates the oil market, 1 for the Brent crude oil prices, while the 2 show the WTI crude oil prices. However, l_i^j show the long-term fluctuations (the secular trend

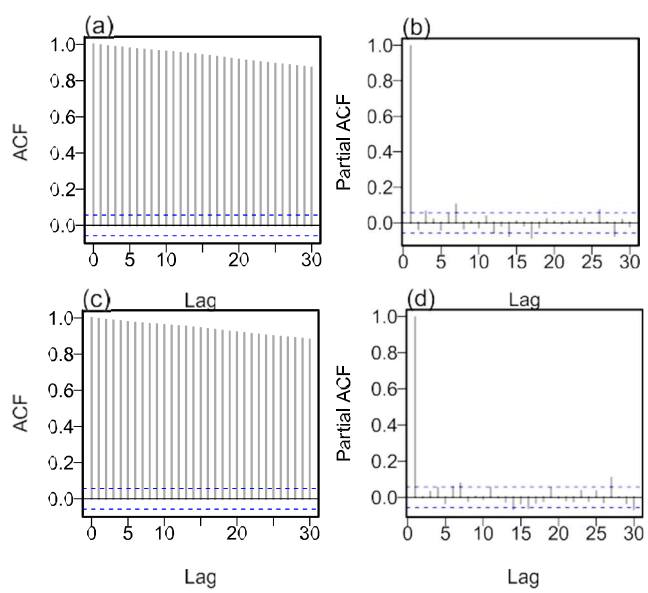

FIGURE 2. Correlogram for both crude oil prices: Brent Oil (a,b) and WTI Oil (c,d).

component). For modeling, the l_i^j component is a function of time i. It is modeled and estimated using four regression models: polynomial regression model, spike regression, smoothing spline regression, and lowess regression model. Therefore, these methods are described in more detail in the following subsection.

B. POLYNOMIAL REGRESSION MODEL

A polynomial regression statistical technique illustrates the causal connection of a single dependent and independent variable. Predictive analysis in machine learning and data science utilizes this technique for parameter estimation. The least squares method is frequently used to fit models for linear regression [51], [52]. Mathematically, the model can be written as.

$$Y_z = \alpha + \beta(z) + \epsilon \tag{2}$$

C. REGRESSION SPLINE MODEL

An approach called spline regression explains non-linear functions and measures non-linearity in the predictor variables. When the hypothesis of linearity may fail or when a confounding variable deviates from the linearity assumption, resulting in residual confounding, this approach is employed [53], [54]. Mathematically, the function of the Regression Spline can be represented as:

$$f(z) = \beta_0 + \beta_1 z + \beta_2 z^2 + \cdots + \beta_p z^p + \sum_{j=1}^{k} \gamma_j (z - \tau_j)^p_+ \tag{3}$$

D. SMOOTHING SPLINES MODEL

Smoothing splines effectively determine functional relationships between a response (Z) and a predictor (X). Smoothing splines can be used effectively in noisy data. Alternatively, spline characteristics are forecasted utilizing the least-squares (LS) penalty term to restrict the sum of squares, which will satisfy the requirements for resolving the knot regions [55], [56].

The SS minimizes the RSS function given by:

$$\text{O.F} = \sum_{i=1}^{n} (z_i - f(x_i))^2 + \lambda \int_{-\infty}^{\infty} f''(x)^2 \, dx \tag{4}$$

E. LOWESS REGRESSION MODEL

In layman's terms, Locally Weighted Scatter plot Smoothing (LOWESS) is a regression analysis technique that draws a smooth line over a scatter plot [57]. This line offers a way to determine the associations between several variables. This line also aids in our understanding of changeable trends. The benefits of (LOWESS) include: i) data is illustrated in a very adaptable way. ii) It is simple to operate. and iii) It can display data trends. Drawbacks of (LOWESS) include complex computations. ii) Not as intuitive as parametric smoothers. iii) Explaining the acquired results is difficult. [58].

To graphically demonstrate the regression models' performance in the figures 3 (a-h). In each sub-figure, actual crude oil prices time series (sky blue–the Brent oil, and pink–the WTI oil) with superimposed estimated non-linear secular trend long-run series using (a, e) Polynomial Regression, (b, f) Regression Spline, (c, g) Smoothing Splines, and (d, h) LOWESS Regression. The figures show that the regression models capture the long-run secular nonlinear trend component well.

Once the long-term fluctuations (the nonlinear trend component have been modeled and estimated), this work proceeds with modeling and estimating the short-run fluctuations (s_i^j). First, it has been obtained in the following way; $s_i^j = (\log(O_i^j) - l_i^j)$. Second, it is modeled and estimated by eight different forecasting methods: four benchmark TS (linear and non-linear AR, ARMA, ESM) and four ML (NNAR, RF, SVR with polynomial and radial basal functions) models described below.

1) AUTO-REGRESSIVE MODEL

A time-series framework called the auto-regressive (AR) model makes projections about future values utilizing past data. An auto-regressive process of order p is defined as a process z_t, if

$$z_t = \alpha_0 + \alpha_1 z_{t-1} + \alpha_2 z_{t-2} + \cdots + \alpha_p z_{t-p} + \epsilon \tag{5}$$

where α_0 $(j = 1 \cdots p)$ are constants and z_t represents random process with 0 mean and variance σ^2 [59]. To estimate the model parameters, this investigation utilizes the ML approach [60]. Lags are incorporated into the model because of their noteworthy findings in the correlogram of the series [61].

2) AUTO-REGRESSIVE INTEGRATED MOVING AVERAGE (ARIMA) MODEL

The Auto-Regressive Integrated Moving Average (ARIMA) model is a kind of regression model that assesses the strength of a single dependent variable in accordance with other changing variables [62], [63].

This three-component model is named AR because it has a variable that regresses on its own historical values and changes over time. d represents the actual observational differencing that stabilizes the series. With lagged observations, the MA model considers the relationship between the actual time series and estimated residual errors [64]. The optimal parameters of ARIMA models are estimates by (MLE) equations [63]. Mathematically, the ARIMA model is demonstrated as

$$\varphi_p(B)(1 - B)^d z_t = \theta_q(B)\epsilon_t \tag{6}$$

3) EXPONENTIAL SMOOTHING AND TBATS MODEL

In 1956, [65] proposed exponential smoothing for the first time in the statistical literature, then expanded by [66].

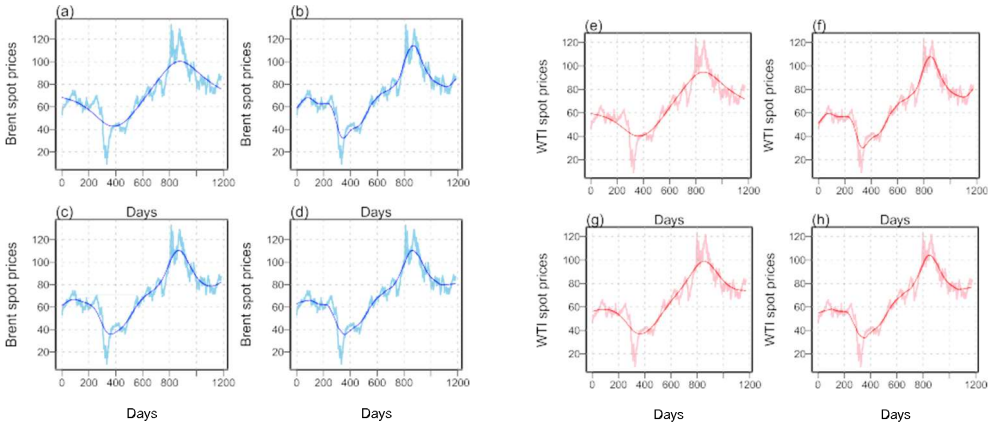

FIGURE 3. Actual crude oil prices time series (sky blue–the Brent oil, and pink–the WTI oil) with superimposed estimated non-linear secular trend long-run series using (a, e) Polynomial Regression, (b, f) Regression Spline, (c, g) Smoothing Splines, and (d, h) LOWESS Regression.

A forecasting technique called TBATS is used to model time series data. Forecasting time series with intricate seasonal patterns by exponential smoothing is the primary goal of this [67].

Mathematically, the ES algorithm in its simplest form is presented by

$$\begin{aligned} z_t &= \ell_{t-1} + \varphi d_{t-1} + \sum_{i=1}^{\times}\sum_{j=1}^{\times} s_{i,j,t} + \varepsilon_t, \\ \ell_t &= \ell_{t-1} + \varphi d_{t-1} + \alpha\varepsilon_t, \\ d_t &= d_{t-1} + \beta\varepsilon_t, \\ s_{i,j,t} &= s_{i,j,t-m_i} + \gamma_{i,j}\varepsilon_t, \\ \varepsilon_t &\sim \mathrm{ARMA}(p, q), \end{aligned}$$

4) HOLT-WINTERS SMOOTHING MODEL

The Holt-Winters model represents time series behavior. A model is usually necessary for forecasting, and Holt-Winters is a method for modeling three components of the time series: a cyclical repeating pattern (seasonality), a slope (trend) throughout time, and a typical value (average) [67]. A mathematical model can be presented as

Additive Model:

$$\begin{aligned} \ell_t &= \alpha(z_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}), \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}, \\ s_t &= \gamma(z_t - \ell_t) + (1-\gamma)s_{t-m}, \\ \hat{z}_{t+h} &= \ell_t + hb_t + s_{t+h-m(k+1)}, \end{aligned}$$

Multiplicative Model:

$$\begin{aligned} \ell_t &= \alpha\frac{y_t}{s_{t-m}} + (1-\alpha)(\ell_{t-1} + b_{t-1}), \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1}, \\ s_t &= \gamma\frac{z_t}{\ell_t} + (1-\gamma)s_{t-m}, \\ \hat{z}_{t+h} &= (\ell_t + hb_t)s_{t+h-m(k+1)}, \end{aligned}$$

5) MULTILAYER PERCEPTRONS MODEL

Multilayer perceptrons are an effective and adaptable technique for time series forecasting. MLPs are a useful tool in a variety of forecasting applications because they can identify intricate patterns and non-linear connections in time series data by utilizing neural network capabilities [68]. Mathematically, a model can be defined as

$$\begin{aligned} \mathbf{h} &= \sigma(\mathbf{W}^{(1)}\mathbf{z} + \mathbf{b}^{(1)}), \\ \hat{\mathbf{y}} &= \varphi(\mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)}), \end{aligned}$$

6) DECISION TREE MODEL

A decision tree is a machine learning model used in time series forecasting that makes predictions about the future values of a time series using a structure resembling a tree. Based on particular characteristics or circumstances, it evaluates the data to find patterns and linkages before making judgments at each node. Particularly useful for capturing complicated patterns and non-linear correlations, especially in non-stationary time series, the procedure generates a projected value at the leaf node [26]. Mathematically, the model can be represented as

$$f(\mathbf{z}) = \begin{cases} c_1 & \text{if } \mathbf{z} \in R_1, \\ c_2 & \text{if } \mathbf{z} \in R_2, \\ \vdots & \\ c_k & \text{if } \mathbf{z} \in R_k, \end{cases} \tag{7}$$

7) NON-PARAMETRIC AUTO-REGRESSIVE MODEL (NPAR)

An additive framework (NPAR), which is an outcome of the non-parametric version of the AR process, facilitates any non-linearity because the relationship involving ψ_z and its previous terms lacks a defined parametric form. Mathematically, it can be represented as

$$\psi_z = f_1(\psi_{z-1}) + f_2(\psi_{z-2}) + \cdots + f_z(\psi_{z-n}) + \epsilon \tag{8}$$

whereas the relation between ψ_z and its prior values is indicated by the smoothing functions and represented by $f_j(j = 1, 2, \cdots, n)$ [69]. Using versatile methods like polynomial functions or kernel regression, this model demonstrates an alternative to conventional parametric models [70].

8) THE AUTO-REGRESSIVE NEURAL NETWORK MODEL (ANN)

An artificial intelligence approach (ANN) predicts future time series values utilizing historical data. A mathematical function, represented by the notation

$$z_t = (\psi_{z-1}) + (\psi_{z-2}) + \cdots + (\psi_{z-n}) \tag{9}$$

where the time delay parameter is indicated by n, is used by the model to account for the historical values of the time series [71]. In order to minimize the discrepancy between predicted and actual values, the ANN model is developed via gradient descent and back-propagation (BP). The ANN model forecasts by first figuring out the AR order, which is trained on a data set, and then figuring out the ideal number of hidden nodes. In order to prevent overfitting the model, significant attention must be paid to the number of training iterations [72]. This research makes use of a standard ANN conception with one hidden layer and three nodes.

9) THE RANDOM FOREST (RF) MODEL

An ensemble of many Regression Trees makes up a random forest. The train data set is selected randomly in the random forest model to generate multiple new data sets bootstrapping that have the identical size as the original. A randomly chosen subset of features may also be included in every new data set. Next, each compilation of data is used to train a separate regression tree, which is then utilized to provide forecasts for new data points. The random forest forecast is the mean of these predictions. Regression trees with varying access to information result in effectively diversified forecasts and are less susceptible to extreme values and biases [73]. However, the Random Forest model is mathematically represented as:

$$X_t = z_{t-1} + z_{t-2} + \cdots + z_{t-p} + \epsilon \tag{10}$$

Provided with a time series z_t, generate a data set of pairs of inputs and outputs (X, z) in which z is the value to be predicted, and X is a vector of the series (lagged values) Here, the total number of lag observations is denoted by p [68].

10) SUPPORT VECTOR REGRESSION (SVR) WITH DIFFERENT KERNELS

SVM theory served as the basis for the first introduction and development of SVR [74]. Regression tasks frequently use the popular machine learning algorithm SVR. Using an unknown function, it converts the nonlinear relation between input vectors and the corresponding actual responses into a straight line in a higher-dimensional feature array [75].

a: RBF KERNEL

The definition of RBF kernel function is [76]:

$$K(z_i, z_j) = \exp\left(-\frac{\|z_i - z_j\|^2}{2\sigma^2}\right) \tag{11}$$

b: PREDICTION FUNCTION

Prediction function (RBF) $f(z)$ is:

$$f(z) = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) K(z_i, z) + b \tag{12}$$

c: POLYNOMIAL KERNEL

Definition of the polynomial kernel function [77]:

$$K(z_i, z_j) = (\gamma z_i \cdot z_j + r)^d \tag{13}$$

where $\gamma > 0$ denotes scaling factor, $r \geq 0$ denotes constant term, and d denotes degree of the polynomial.

d: PREDICTION FUNCTION

Prediction function of (PFK) $f(z)$ is:

$$f(z) = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) K(z_i, z) + b \tag{14}$$

F. ACCURACY MEASURES

We employed seven unique assessment indicators to evaluate the model from multiple perspectives. Among these measurements are the Directional Statistic (DS), Pearson Correlation Coefficient (PCC), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). Equations 15, 16, 17, and 18 provide the mathematical formulas for assessing forecast accuracy, accordingly.

$$\mathrm{MAE} = \frac{1}{n} \sum_{t=1}^{n} |Z_i - \hat{Z}_i|, \tag{15}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (Z_i - \hat{Z}_i)^2}, \tag{16}$$

$$\mathrm{PCC} = \mathrm{Correlation}\ Z_i, \hat{Z}_i, \tag{17}$$

where n is the number of observations in the data set, and Z_i and $\hat{Z}_i$ are the t_{th} estimated and observed data points, respectively.

The DS is used to evaluate the ability of direction prediction, which is defined as follows:

$$\mathrm{DS} = \frac{1}{n} \sum_{i=1}^{n} \beta_i,$$

$$\beta_i = 1, (Z_{i+1} - Z_i) \times (\hat{Z}_{i+1} - Z_i) \geq 0, \quad \text{otherwise} \tag{18}$$

Higher CC and DS values, when combined with lower MAE and RMSE values, signify more accurate forecasting and better direction prediction.

As within the proposed hybrid time series forecasting technique, this work used four different estimation methods

for the non-linear long-run secular trend component, whereas the forecasting of the rest component (residual part) was by eight different forecasting models: four standard time series and four standard machine learning models. Hence, estimating trend and residual components led us to compare thirty-two other hybrid models. For easy to read, in Table 2 this work represents each hybrid model in the following way: the first two letters show the trend component estimation method. In comparison, the last four letters indicate the residual component estimation model. For instance, PRARMA denotes PR (polynomial regression model)–estimation for trend component and ARMA (autoregressive moving averages model)–residual part. Similarly, SR denotes (short-run), LR denotes (long-run), and SS denotes the (smoothing spline). Thus, the highest possible daily spot price estimate for (Brent and WTI) oil is produced by combining the separate forecast models.

III. DAILY CRUDE OIL PRICES FORECASTING RESULTS

This investigation aims to efficiently forecast the daily market prices for WTI and Brent oil (1200 data points) for both markets. Training (80%) and testing (20%) data sets are separated for both markets, and proposed models are applied and evaluated accordingly. As exploratory data analysis is a potent technique that visually reveals the features of data sets, this research displays the data set in Figure 1 for both markets to determine whether nonlinear trends, seasonality, non-normality, and non-stationarity are present. From Figures 1 (a, b), due to non-linear behavior, there seems to be a non-stationarity for both oil markets, which is confirmed by applying the (ADF) test. Figure 1 (b) illustrates how WTI market prices changed during the study period, fluctuating between about $20 to $120. Furthermore, there seem to be events of both significant price jumps and declines, indicating a high degree of volatility. The identical variables that determine WTI prices probably also influence the volatility of the Brent spot prices, which display substantial spikes and declines.

The features of the data before and after several modifications, such as differencing (to investigate stationary behavior) and taking the natural logarithm (to address volatility stabilization), are summarized in Table 1, and an overview of a data set distribution structure, variability, and central pattern can be obtained using descriptive statistics. Table 1 shows the lowest recorded price for Brent oil is 9.12, the highest price is 133.18, Q1 is 59.46, Q2 is 83.53, and the median, mean, and mode are 70.97, 70.71, 485.45, and 3.20, respectively. The distribution of Brent oil may not be stationary, as indicated by the (ADF) statistic of -1.74, which provides proof against the existence of a unit root, so we apply the first difference of the Brent oil series. Additionally, to determine if the series has a normal distribution, the Jarque-Bera (JB) is employed, which results in 4.87, suggesting that the data distribution is nearly normal. Additionally, the minimal log-transformed Brent oil value is 2.21, while the Q1 price is 4.09, and the median, mean, and mode are 4.25, 4.21, and 0.13, respectively. With a skew value of -1.29, Brent oil prices skewed to the left; the maximum is 4.89, Q3 is 4.43, and the kurtosis is 0.67. The log transformation leads to enhanced stationarity, as indicated by the ADF statistic of -2.20; log-transformed prices may not be normally distributed, as suggested by the significant departure from normality indicated by the (JB) estimate of 791.53.

The data set displayed in table 1 shows a minimum price of 8.91 for WTI Oil, with Q1 being 53.39, median being 65.03, mean being 66.40, mode being 78.32, variance being 441.92, skewness being 0.17, kurtosis being 2.98, Q3 being 79.60, and Maximum being 123.64. The ADF statistic of -1.68 suggests that the time series may not be stationary and requires differencing, while the (JB) value of 5.80 suggests that the distribution of prices is near normal. In terms of log-transformed pricing, the lowest log-transformed WTI price is 2.19, the highest is 4.82, the third quartile is 4.38, the mean is 4.14, the variance is 0.13, the skewness is -1.17, and the mode is 4.36. With the log transformation, there is moderate evidence to exclude the existence of a unit root, according to the ADF statistic of -2.33. Significant deviation from normality is indicated by the (J-B) statistic of 639.51. The logarithm transformation provides distributions that greatly depart from normality in the two scenarios, i.e. (WTI and Brent), but it additionally assists in stabilizing variability and enhancing uniformity.

TABLE 1. Descriptive statistics.

Market	Brent Oil		WTI Oil	
Statistic	Original Prices	Log Prices	Original Prices	Log Prices
Minimum	9.12	2.21	8.91	2.19
Q1	59.46	4.09	53.39	3.98
Median	70.34	4.25	65.03	4.17
Mean	70.97	4.21	66.40	4.14
mode	70.71	4.26	78.32	4.36
Variance	485.45	0.13	441.92	0.13
Skewness	0.12	-1.29	0.17	-1.17
Kurtosis	3.20	6.07	2.98	5.76
Q3	83.53	4.43	79.60	4.38
Maximum	133.18	4.89	123.64	4.82
ADF-Statistic	-1.74	-2.20	-1.68	-2.33
JQ-Statistic	4.87	791.53	5.80	639.51

The procedures listed below must be taken in order to use the proposed hybrid time series forecasting technique described in Section II to provide daily forecasts for Brent and WTI crude oil: Prior to normalizing the data, handling missing values, and stabilizing variance, the original crude oil price time series must be treated. Second, it divides the purified crude oil price time series into two major parts: the nonlinear long-term trend (long-run fluctuations) and the residual (short-term fluctuations). The nonlinear secular trend series is modeled using various regression models, including linear, spline, lowess, and smoothing spline regression models. On the other hand, the short-run random part is modeled and forecasted using four standard time series (linear and nonlinear autoregressive, autoregressive moving average, exponential smoothing models) and four machine learning (Neural network autoregressive, Random

forecast, support vector regression with polynomial and radial basal functions) models. The forecast from both parts is summed to obtain the final daily forecasts. However, various performance criteria (mean errors, correlation coefficient, and directional statistics) are used to verify the efficiency of the proposed hybrid time-series framework. There are possibly thirty-two hybrid models within the proposed hybrid time series forecasting technique. For these thirty-two hybrid models, the daily out-of-sample forecast accuracy measures (MAE, RMSE, PCC, and DS) are tabulated in Table 3. Before discussing the results table, it is worth mentioning that the smaller the accuracy mean errors (MAE and RMSE) and the greater the value of the coefficient of correlation and directional statistic (PCC and DS), the higher the forecasting efficiency and accuracy model.

The proposed hybrid models are employed to project the Brent and WTI oil prices, shown in table 2 along with Mean Error metrics. The model that performs the best in terms of DS, correlation, and lowest error measures for Brent oil found is LRNPAR with MAE of 1.28106 ↓, RMSE of 1.59259 ↓, PPC of 0.94158 ↑, and DS of 0.82149 ↑. Additionally, the PRARMA model results in less error w.r.t MAE of 1.28299 followed by SSSVRP with MAE of 1.28956, and the fourth proposed model with less MAE is SRSVRP with the value of 1.28987 ↓.

Moreover, for the WTI oil prices, the model LRNPAR model outputs the lowest MAE of 1.25730 ↓, RMSE of 1.55390 ↓, PCC of 0.93890 ↑, DS of 0.82320 ↑. SRSVRP, which results in the second lowest MAE of 1.26648↓ and SSSVRP of 1.26679 ↓ and lastly, the PRAR model results in MSE of 1.26995 ↓, furthermore, the rest of the combinations of proposed models are given in table 2.

Graphical assessment of the best combinations of the proposed model is presented in figure 5, the actual and projected prices for WTI and Brent oil based on various forecasting models. It should be noted that the PRARMA model seems to be closely matching the accurate Brent prices, suggesting that its forecasts seem entirely accurate; with a slight departure from the actual prices, the SRSVRP and SSSVRP models equally demonstrate exemplary performance. Comparatively to the other models, LRNPAR has somewhat more deviation from the real prices, suggesting a minor decrease in accuracy. Similarly, the PRAR approach exhibits good accuracy in prediction and closely gauges the real WTI market prices. With projections that nearly resemble real values, SRSVRP and SSSVRP likewise perform well. Despite capturing the general trend, LRNPAR displays considerable discrepancies from actual WTI oil prices. Additionally, the correlation plot for the observed (testing) versus the fitted values is given in Figure 4 for the four best-proposed forecasting models for Brent and WTI oil prices.

IV. DISCUSSION

In this section, we will comprehensively compare the best crude oil forecasting model in this work to baseline models and the lecture's best crude oil forecasting models. In addition, we will discuss the future direction and recommendations for policymakers. Thus, the following subsection gives more details on the comparison with baseline models, the best forecasting model in the literature, future studies, and policy recommendations.

A. COMPARISON WITH BASELINE MODELS

Table 3 shows the outcomes of different forecasting approaches for Brent and WTI oil using machine learning and classical time series models. According to the AR Model results, the WTI oil has an MAE of 1.2783, an RMSE of 1.5811, Pearson's correlation coefficient of 0.9372, and a directional symmetry (DS) of 0.7805. The results for Brent oil indicate that the MAE is 1.4564, RMSE is 1.9508, PCC is 0.9152, and DS is 0.7317. According to the AR model, both markets have a strong linear association, and they perform reasonably well with a high PCC. The moderate MAE and RMSE values indicate reasonable accuracy. The predictive model can anticipate the pattern of fluctuations in prices quite accurately, according to the DS value. With PCC values of 0.9397 and 0.937, respectively, this suggests a strong correlation for both markets. The TBATS outperforms AR in terms of MAE of 1.2996 and RMSE of 1.6299 for Brent oil. The DS for WTI oil is notably high at 0.811, suggesting that TBATS can be more advantageous for determining the pattern of WTI oil price fluctuations than other sixty-one combinations of models. For Brent and WTI oils, NPAR performs well with low MAE and RMSE values of 1.2924, 1.6139, 1.2707, and 1.5544, respectively, and high PCC values. In both markets, ARIMA surpasses NPAR, exhibiting a significant correlation and accurate direction. With slightly larger errors but maintaining significant correlations and credible directional accuracy, HW performs comparably to ARIMA and NPAR. Theta M performs efficiently, producing outcomes comparable to those of HW, ARIMA, and NPAR, suggesting that it constitutes a valid approach for both markets. On the other hand, compared with various models in Table 3, the MLP exhibits higher erroneous and low PCC values but retains directional solid accuracy. SVRP demonstrates moderate deviations and correlations and performs comparably to MLP. Concerning MAE and RMSE for both Brent and WTI oil, SVRL performs somewhat better while preserving strong correlations. In comparison to the other models, SVRLP performs substantially worse, exhibiting notably lower correlations and larger errors, particularly for WTI Oil. With strong correlation and modest errors, SVRB performs well; nonetheless, its directional accuracy is marginally less than that of other models. Compared to certain other models, RF performs well, although having slightly larger residuals and low PCC values. With its largest errors and lowest correlations, DT performs poorly out of all the models. Based on each of the metrics (MAE, RMSE, PCC, and DS) for Brent Oil, the models that have the highest accuracy are as follows: As per the Table 3, the ARIMA model has the lowest mean absolute error (MAE) of 1.2921 out of all the

TABLE 2. Accuracy metrics.

Market	Brent Oil				WTI Oil			
Model	MAE	RMSE	PCC	DS	MAE	RMSE	PCC	DS
PRAR	1.29231	1.62236	0.93980	0.78659	1.26995	1.58978	0.93664	0.80488
PRNPAR	1.30241	1.68273	0.93909	0.78659	1.33013	1.68766	0.93376	0.79878
PRARMA	1.28299	1.61274	0.94086	0.81098	1.27167	1.59066	0.93669	0.80488
PRES	1.29730	1.62913	0.93956	0.78749	1.27270	1.58234	0.93713	0.78049
PRANN	1.36611	1.77855	0.93007	0.77439	1.35246	1.69026	0.92981	0.78049
PRSVRP	1.28512	1.61417	0.94000	0.79268	1.27784	1.56900	0.93822	0.80488
PRSVRB	1.35566	1.67878	0.93878	0.78659	1.33265	1.63087	0.93483	0.77439
PRRF	1.44090	1.99613	0.91376	0.71951	1.48220	1.89686	0.90861	0.75610
SRAR	1.30751	1.63554	0.93915	0.78049	1.30736	1.59867	0.93644	0.80488
SRNPAR	1.29418	1.60364	0.94114	0.78659	1.27207	1.55441	0.93949	0.80488
SRARMA	1.30101	1.62659	0.94016	0.77439	1.31132	1.59909	0.93646	0.81098
SRES	1.31113	1.63940	0.93910	0.77439	1.29245	1.58792	0.93714	0.78049
SRANN	1.42389	1.78978	0.92802	0.75610	1.32211	1.66389	0.93150	0.78659
SRSVRP	1.28987	1.61515	0.93967	0.78659	1.26648	1.55808	0.93874	0.79878
SRSVRB	1.31302	1.63309	0.93929	0.78049	1.28397	1.56978	0.93908	0.78659
SRRF	1.41993	1.76608	0.92958	0.75000	1.32412	1.65124	0.93280	0.76829
SSAR	1.30763	1.63589	0.93911	0.78049	1.30718	1.59865	0.93642	0.80488
SSNPAR	1.29266	1.60390	0.94105	0.81259	1.27116	1.55400	0.93946	0.80488
SSARMA	1.29568	1.61962	0.94058	0.78049	1.31072	1.59885	0.93646	0.81098
SSES	1.31117	1.63965	0.93908	0.77439	1.29232	1.58781	0.93715	0.78049
SSANN	1.40724	1.76585	0.93005	0.77439	1.32572	1.67427	0.93039	0.78659
SSSVRP	1.28956	1.61456	0.93971	0.78049	1.26679	1.55857	0.93872	0.80488
SSSVRB	1.31049	1.63169	0.93931	0.78049	1.28347	1.56886	0.93913	0.78659
SSRF	1.35960	1.72722	0.93247	0.74390	1.35912	1.69455	0.92774	0.78049
LRAR	1.30610	1.63259	0.93931	0.78049	1.30361	1.59375	0.93670	0.80488
LRNPAR	1.28106 ↓	1.59259 ↓	0.94158 ↑	0.82149↑	1.25730 ↓	1.55390 ↓	0.93890 ↑	0.82320 ↑
LRARMA	1.29605	1.61873	0.94060	0.77439	1.30735	1.59448	0.93669	0.81098
LRES	1.31086	1.63729	0.93919	0.78049	1.29096	1.58609	0.93724	0.78049
LRANN	1.41773	1.78212	0.92784	0.76829	1.32189	1.67058	0.93102	0.80488
LRSVRP	1.28764	1.61012	0.93984	0.78049	1.26343	1.55522	0.93888	0.79878
LRSVRB	1.30314	1.62376	0.93944	0.78049	1.28154	1.56424	0.93932	0.79268
LRRF	1.47570	1.82683	0.92312	0.73171	1.38755	1.71807	0.92695	0.76829

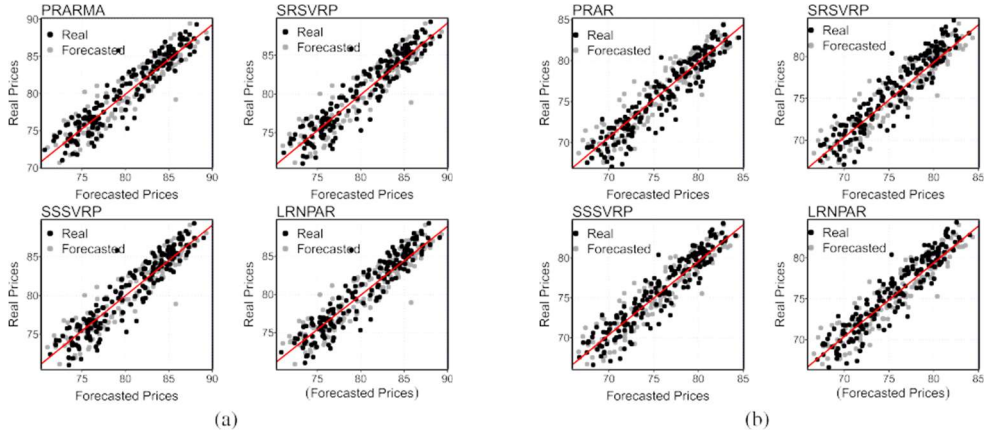

FIGURE 4. Correlation plots of best performance model for Brent (a) and WTI oil (b).

models. The NPAR model has the lowest RMSE value of 1.6139, marginally lower than ARIMA. The highest PCC that the ARIMA model can achieve is 0.9405. Finally, the three models (i.e., HW, Theta M, SVRP) accurately capture the pattern of Brent oil with a coefficient value of 0.7927. In the same manner, the NPAR model exhibits the lowest MAE of 1.2721, RMSE of 1.5544, and greatest PCC value of 0.9395. Conversely, the TBATS model, with a coefficient value of 0.811, slightly higher than NPAR, accurately represents the WTI oil pattern for the DS value. When comparing the baseline models in Table 3 and the Proposed hybrid model LRNPAR in Table 2, a number of measures, such as (MAE), (RMSE), (PCC), and (DS), demonstrate a notable improvement as the values for both Brent and WTI oil market with the values of 1.28106, 1.59259, 0.94158, 0.82149 and 1.25730, 1.55390, 0.93890, 0.82320, respectively. The hybrid models make use of a hybrid strategy that blends the advantages of several approaches,

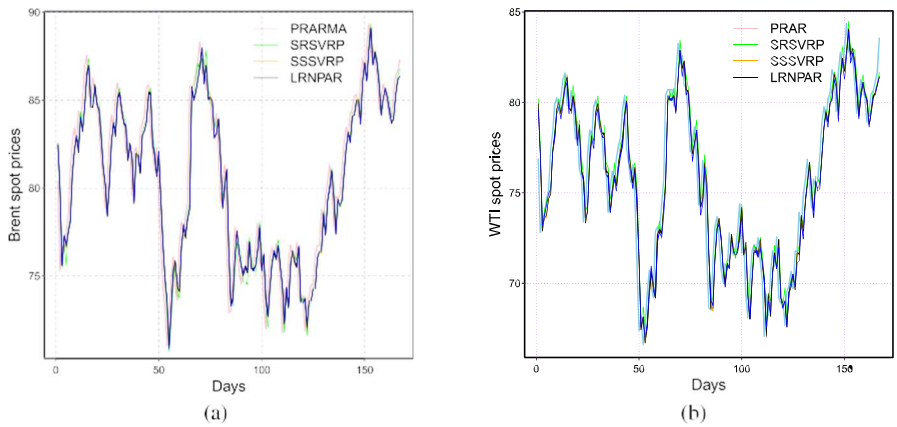

FIGURE 5. Observed versus fitted time series plot for Brent (a) and WTI (b) oil.

including neural networks, machine learning, and statistical techniques. This combination improves forecast accuracy and robustness by addressing the shortcomings of individual models. Baseline models such as ARIMA and TBATS may miss complex patterns in real-world data, which are limited by linearity assumptions or particular time series features. On the other hand, hybrid models improve generalization across a variety of datasets by adjusting to seasonality and non-linear patterns. The outcomes demonstrate how useful hybrid models are for time series forecasting in volatile markets such as the oil industry. By surpassing standard models, the proposed methods provide practitioners with a viable tool for making accurate and predictable projections.

B. COMPARISON WITH THE LITERATURE BEST MODELS

Compared with the [2] which focuses on a comprehensive study of the forecasting of Brent crude oil prices through the assessment of decomposed hybrid combinations of models, the present study improves the results in terms of accuracy prediction when the training data set to increase from 50% to 75% and 90%.

In another study, crude oil prices were calculated down by [78] using the Ensemble Empirical Mode Decomposition (EEMD) approach into a series of independent Intrinsic Mode Functions (IMFs) and a residual term. This study uses four different methods to calculate the MAE and RMSE. For the daily crude oil, MAE and RMSE were 4.76 and 4.78, accordingly but when compared to the current study, the proposed framework improves accuracy significantly, lowering MAE by around 73% and RMSE by about 67% when compared to the baseline.

In order to evaluate its predicting performance with the current AdaBoost-LSTM model, [79] this study suggests a hybrid model that combines the AdaBoost algorithm with GRU. Compared to the current proposed study the improvements in MAE are LSTM: 61.15%, GRU: 57.86%, AdaBoost-LSTM: 21.83%, AdaBoost-GRU: 9.63% and in terms of RMSE LSTM: 71.15%, GRU: 69.76%, AdaBoost-LSTM: 47.29%, AdaBoost-GRU: 35.38%.

Reference [80] This study used the BOP dataset to create an effective methodology for predicting Brent oil prices using the Bi-LSTM network. The current study achieved significant improvements in terms of MAE: 56.91% improvement over CNN-LSTM, 52.42% improvement over CNN-Bi-LSTM, and 24.26% improvement over LSTM. For RMSE, the proposed model in this study achieves 57.60% improvement over CNN-LSTM, 54.05% improvement over CNN-Bi-LSTM, and 25.35% improvement over LSTM. In comparison to CNN-LSTM, CNN-Bi-LSTM, and LSTM, the proposed approach exhibits noteworthy improvements in MAE and RMSE.

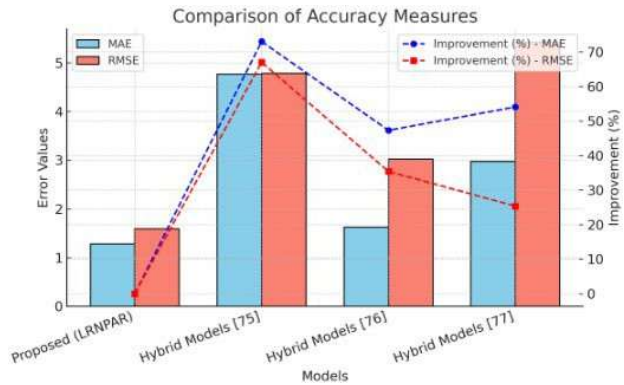

FIGURE 6. Accuracy measures of the proposed models versus those presented in the literature.

Figure 6 illustrates a comparison of the accuracy metrics between the proposed LRNPAR model and existing hybrid models found in the literature, focusing on Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The LRNPAR model demonstrates the lowest error rates (MAE = 1.28, RMSE = 1.59), highlighting its superior effectiveness. In contrast, Hybrid Models [74], [75], and [76] show considerably higher error rates, indicating a reduction in

TABLE 3. The performance comparison of baseline models and the proposed hybrid forecasting model.

Market	Brent Oil				WTI Oil			
Model	MAE	RMSE	PCC	DS	MAE	RMSE	PCC	DS
AR	1.4564	1.9508	0.9152	0.7317	1.2783	1.5811	0.9372	0.7805
TBATS	1.2996	1.6299	0.9397	0.7622	1.2916	1.5852	0.937	0.811
NPAR	1.2924	1.6139	0.9402	0.7866	1.27207	1.55441	0.93949	0.80488
ARIMA	1.2921	1.6155	0.9405	0.7805	1.2961	1.5911	0.9366	0.7866
HW	1.3008	1.6284	0.9396	0.7927	1.2783	1.5811	0.9372	0.7805
ThetaM	1.2989	1.6285	0.9396	0.7927	1.2753	1.5814	0.9371	0.7805
MLP	1.3334	1.6959	0.9349	0.7744	1.3022	1.6381	0.933	0.7866
SVRP	1.3233	1.6425	0.9394	0.7927	1.2957	1.5877	0.937	0.7805
SVRL	1.3235	1.643	0.9393	0.7866	1.2962	1.5879	0.937	0.7805
SVRLP	2.222	3.0807	0.8633	0.7561	1.8787	2.6093	0.8838	0.7744
SVRB	1.3822	1.755	0.9351	0.7683	1.3683	1.7113	0.9314	0.7561
RF	1.3955	1.7812	0.9265	0.7561	1.3475	1.6953	0.9261	0.7622
DT	3.466	4.2582	0.8297	0.5915	3.5472	4.2295	0.7646	0.6037
Proposed - LRNPAR	1.28106 ↓	1.59259 ↓	0.94158 ↑	0.82149↑	1.25730 ↓	1.55390 ↓	0.93890 ↑	0.82320 ↑

TABLE 4. Accuracy measures of the proposed models versus those presented in the literature.

Model	MAE	RMSE	Improvement (%)
Proposed(LRNPAR)	1.28	1.59	- - - - -
Hybrid Models [78]	4.76	4.78	73.00 and 67.00
Hybrid Models [79]	1.62	3.02	47.29 and 35.38
Hybrid Models [80]	2.97	5.42	54.05 and 25.35

accuracy. The percentage improvements in MAE and RMSE indicate that LRNPAR surpasses the hybrid models by decreasing errors by as much as 73% and 67%, respectively. These findings imply that the LRNPAR model offers a more precise and dependable prediction framework than established methods.

C. FUTURE STUDIES

Assuming that PRAR and PRARMA models operate effectively, future studies might concentrate on investigating hybrid models or incorporating cutting-edge artificial intelligence methods. These methods might increase forecasting accuracy, especially when it comes to monitoring abrupt and erratic changes in the oil market, such as price shocks or geopolitical upheavals. Furthermore, adding variables other than conventional price data may improve model forecasts. For instance, a more thorough understanding of the oil market and its factors may be possible by combining supply chain dynamics, economic indicators (such as GDP and inflation rates), and political events (such as policy changes, conflicts, or sanctions).

On the other hand, their computational complexity is An essential factor in assessing machine learning models—especially in tasks related to acoustic recognition. This research would greatly benefit from a more thorough exploration in this domain, particularly when evaluating time-frequency representations like spectrograms and scalograms. As highlighted by Phan [81], the selection of input representation substantially influences both the training and inference durations of models, with scalograms typically causing greater computational demands due to the resource-intensive nature of continuous wavelet transforms (CWT). Moreover, Phan [82] introduces a technique for mitigating CWT complexity through hop-size adjustment, effectively lessening redundancy while maintaining recognition accuracy. Incorporating such strategies or examining the runtime performance of models could offer essential perspectives on the balance between accuracy and efficiency. Future investigations may expand these concepts to enhance resource utilization in real-time or embedded acoustic systems.

D. POLICY RECOMMENDATIONS

When it comes to predicting WTI and Brent crude oil prices, the prediction models put out in this work—PRAR, PRARMA, SRSVRP, and SSSVRP—have proven to be remarkably accurate. These models are applicable in a number of real-world contexts for policymakers and business executives. For example, these models can be used to predict changes in oil prices and how they would affect international trade ties, which can help in trade talks. Their ability to predict long-term price swings can also help with future investment strategies and infrastructure development. By offering insights into the possibility of changes in oil prices that could affect energy markets, these models could aid in developing renewable energy legislation within the environmental policy framework. Furthermore, the models aid in the formation of strategic policies concerning energy security, budgetary decisions, and monetary plans, guaranteeing that government bodies are better equipped to respond to changes in the oil market.

V. CONCLUSION

The present work is an attempt to model and foresee the oil prices of two dominant markets, Brent and WTI. To achieve this end, the current analysis uses the classical time series method, regression method with different techniques, and machine learning models with different kernels. The current analysis explores the basic characteristics of both series by applying the central tendencies, a measure of dispersion, normality, and stationarity test to get a clear picture of

the data. The study suggests that the models based on the regression methods performed more efficiently than other existing methods and models that perform well in projecting crude oil costs under various scenarios can be thoroughly understood by comparing them using metrics such as (MAE), (RMSE), (PCC), and (DS). Additionally, the graphical representation of crude oil price patterns and the study of predictions for WTI and Brent oil prices provide a significant understanding of the efficacy and accuracy of different prediction approaches. The current study focuses on the challenge of foreseeing oil prices and how important it is to choose models with caution that strike the right balance between simplicity, accuracy, and flexibility in response to changes in the oil market. Given the consistent forecasts that may assist the execution of decisions in such a changeable environment, the PRAR framework, in particular, emerges as a robust tool for oil buyers and sellers.

REFERENCES

[1] L.-T. Zhao, Y. Wang, S.-Q. Guo, and G.-R. Zeng, ''A novel method based on numerical fitting for oil price trend forecasting,'' *Appl. Energy*, vol. 220, pp. 154–163, Jun. 2018.

[2] H. Iftikhar, A. Zafar, J. E. Turpo-Chaparro, P. C. Rodrigues, and J. L. López-Gonzales, ''Forecasting day-ahead Brent crude oil prices using hybrid combinations of time series models,'' *Mathematics*, vol. 11, no. 16, p. 3548, Aug. 2023.

[3] K. Ma, Y. Yu, B. Yang, and J. Yang, ''Demand-side energy management considering price oscillations for residential building heating and ventilation systems,'' *IEEE Trans. Ind. Informat.*, vol. 15, no. 8, pp. 4742–4752, Aug. 2019.

[4] F. Jan, H. Iftikhar, M. Tahir, and M. Khan, ''Forecasting day-ahead electric power prices with functional data analysis,'' *Frontiers Energy Res.*, vol. 13, Mar. 2025, Art. no. 1477248.

[5] A. Safari and M. Davallou, ''Oil price forecasting using a hybrid model,'' *Energy*, vol. 148, pp. 49–58, Apr. 2018.

[6] X. Zhang, X. Yang, and Q. He, ''Multi-scale systemic risk and spillover networks of commodity markets in the bullish and bearish regimes,'' *North Amer. J. Econ. Finance*, vol. 62, Nov. 2022, Art. no. 101766.

[7] C.-Q. Huang, Q.-H. Huang, X. Huang, H. Wang, M. Li, K.-J. Lin, and Y. Chang, ''XKT: Toward explainable knowledge tracing model with cognitive learning theories for questions of multiple knowledge concepts,'' *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 7308–7325, Nov. 2024.

[8] H. Iftikhar, F. Khan, E. A. Torres Armas, P. C. Rodrigues, and J. L. López-Gonzales, ''A novel hybrid framework for forecasting stock indices based on the nonlinear time series models,'' *Comput. Statist.*, vol. 2025, pp. 1–24, Mar. 2025.

[9] L. Yu, W. Dai, and L. Tang, ''A novel decomposition ensemble model with extended extreme learning machine for crude oil price forecasting,'' *Eng. Appl. Artif. Intell.*, vol. 47, pp. 110–121, Jan. 2016.

[10] M. Hamdi and C. Aloui, ''Forecasting crude oil price using artificial neural networks: A literature survey,'' *Econ. Bull.*, vol. 35, no. 2, pp. 1339–1359, Jun. 2015.

[11] X. Peng, S. Song, X. Zhang, M. Dong, and K. Ota, ''Task offloading for IoAV under extreme weather conditions using dynamic price driven double broad reinforcement learning,'' *IEEE Internet Things J.*, vol. 11, no. 10, pp. 17021–17033, Oct. 2024.

[12] I. Shah, H. Iftikhar, and S. Ali, ''Modeling and forecasting electricity demand and prices: A comparison of alternative approaches,'' *J. Math.*, vol. 2022, no. 1, Jan. 2022, Art. no. 3581037.

[13] Y. Zhu, Y. Zheng, and Z. Ren, ''Household welfare loss from energy price crisis: Evidence from China,'' *Energy Econ.*, vol. 138, Oct. 2024, Art. no. 107836.

[14] M. Qureshi, H. Iftikhar, P. C. Rodrigues, M. Z. Rehman, and S. A. A. Salar, ''Statistical modeling to improve time series forecasting using machine learning, time series, and hybrid models: A case study of Bitcoin price forecasting,'' *Mathematics*, vol. 12, no. 23, p. 3666, Nov. 2024.

[15] H. Iftikhar, M. Khan, J. E. Turpo-Chaparro, P. C. Rodrigues, and J. L. López-Gonzales, ''Forecasting stock prices using a novel filtering-combination technique: Application to the Pakistan stock exchange,'' *AIMS Math.*, vol. 9, no. 2, pp. 3264–3288, 2024.

[16] H. Iftikhar and D. S. Jamil, ''Demand forecasting with a better lstm network in supply chain management though machine learning,'' *Int. J. Innov. Stud.*, vol. 8, no. 1, pp. 1025–1049, 2024.

[17] M. Khashei and M. Bijari, ''A new hybrid methodology for nonlinear time series forecasting,'' *Model. Simul. Eng.*, vol. 2011, no. 1, pp. 1–15, Jan. 2011.

[18] H. Iftikhar, M. Khan, M. S. Khan, and M. Khan, ''Short-term forecasting of monkeypox cases using a novel filtering and combining technique,'' *Diagnostics*, vol. 13, no. 11, p. 1923, May 2023.

[19] E. Wang, Y. Yang, J. Wu, W. Liu, and X. Wang, ''An efficient prediction-based user recruitment for mobile crowdsensing,'' *IEEE Trans. Mobile Comput.*, vol. 17, no. 1, pp. 16–28, Jan. 2018.

[20] H. Iftikhar, F. Khan, P. C. Rodrigues, A. A. Alharbi, and J. Allohibi, ''Forecasting of inflation based on univariate and multivariate time series models: An empirical application,'' *Mathematics*, vol. 13, no. 7, p. 1121, Mar. 2025.

[21] A. Timmermann, *Handbook of Economic Forecasting*. Amsterdam, The Netherlands: Elsevier, 2013.

[22] W. M. Cuba, J. C. Huaman Alfaro, H. Iftikhar, and J. L. López-Gonzales, ''Modeling and analysis of monkeypox outbreak using a new time series ensemble technique,'' *Axioms*, vol. 13, no. 8, p. 554, Aug. 2024.

[23] H. Iftikhar, N. Bibi, P. Canas Rodrigues, and J. L. López-Gonzales, ''Multiple novel decomposition techniques for time series forecasting: Application to monthly forecasting of electricity consumption in Pakistan,'' *Energies*, vol. 16, no. 6, p. 2579, Mar. 2023.

[24] B. Liu, M. Li, Z. Ji, H. Li, and J. Luo, ''Intelligent productivity transformation: Corporate market demand forecasting with the aid of an AI virtual assistant,'' *J. Org. End User Comput.*, vol. 36, no. 1, pp. 1–27, Jan. 2024.

[25] S. M. Gonzales, H. Iftikhar, and J. L. López-Gonzales, ''Analysis and forecasting of electricity prices using an improved time series ensemble approach: An application to the Peruvian electricity market,'' *AIMS Math.*, vol. 9, no. 8, pp. 21952–21971, 2024.

[26] H. Iftikhar, S. M. Gonzales, J. Zywiołek, and J. L. López-Gonzales, ''Electricity demand forecasting using a novel time series ensemble technique,'' *IEEE Access*, vol. 12, pp. 88963–88975, 2024.

[27] N. Carbo-Bustinza, H. Iftikhar, M. Belmonte, R. J. Cabello-Torres, A. R. H. De La Cruz, and J. L. López-Gonzales, ''Short-term forecasting of ozone concentration in metropolitan lima using hybrid combinations of time series models,'' *Appl. Sci.*, vol. 13, no. 18, p. 10514, Sep. 2023.

[28] H. Iftikhar, M. Qureshi, J. Zywiołek, J. L. López-Gonzales, and O. Albalawi, ''Short-term PM2.5 forecasting using a unique ensemble technique for proactive environmental management initiatives,'' *Frontiers Environ. Sci.*, vol. 12, Sep. 2024, Art. no. 1442644.

[29] A. Amano, ''A small forecasting model of the world oil market,'' *J. Policy Model.*, vol. 9, no. 4, pp. 615–635, Dec. 1987.

[30] L. Tang and S. Hammoudeh, ''An empirical exploration of the world oil price under the target zone model,'' *Energy Econ.*, vol. 24, no. 6, pp. 577–596, Nov. 2002.

[31] S. Moshiri and F. Foroutan, ''Forecasting nonlinear crude oil futures prices,'' *Energy J.*, vol. 27, no. 4, pp. 81–96, Oct. 2006.

[32] W. Xie, L. Yu, S. Xu, and S. Wang, ''A new method for crude oil price forecasting based on support vector machines,'' in *Proc. Int. Conf. Comput. Sci.*, May 2006, pp. 444–451.

[33] A. Khashman and N. Nwulu, ''Support vector machines versus back propagation algorithm for oil price prediction,'' in *Proc. Int. Symp. Neural Netw.*, May 2011, pp. 530–538.

[34] J. Chai, L.-M. Xing, X.-Y. Zhou, Z. G. Zhang, and J.-X. Li, ''Forecasting the WTI crude oil price by a hybrid-refined method,'' *Energy Econ.*, vol. 71, pp. 114–127, Mar. 2018.

[35] A. Nademi and Y. Nademi, ''Forecasting crude oil prices by a semiparametric Markov switching model: OPEC, WTI, and Brent cases,'' *Energy Econ.*, vol. 74, pp. 757–766, Aug. 2018.

[36] J. Wu, F. Miu, and T. Li, ''Daily crude oil price forecasting based on improved CEEMDAN, SCA, and RVFL: A case study in WTI oil market,'' *Energies*, vol. 13, no. 7, p. 1852, Apr. 2020.

[37] T. Yao and Y.-J. Zhang, ''Forecasting crude oil prices with the Google index,'' *Energy Proc.*, vol. 105, pp. 3772–3776, May 2017.

[38] P. B. Solibakke, ''Forecasting hourly WTI oil front monthly price volatility densities,'' *Quant. Finance Econ.*, vol. 8, no. 3, pp. 466–501, 2024.

[39] L. Yu, H. Xu, and L. Tang, ''LSSVR ensemble learning with uncertain parameters for crude oil price forecasting,'' *Appl. Soft Comput.*, vol. 56, pp. 692–701, Jul. 2017.

[40] Y. Bao, X. Zhang, L. Yu, K. K. Lai, and S. Wang, ''An integrated model using wavelet decomposition and least squares support vector machines for monthly crude oil prices forecasting,'' *New Math. Natural Comput.*, vol. 7, no. 2, pp. 299–311, May 2011.

[41] K. He, L. Yu, and K. K. Lai, ''Crude oil price analysis and forecasting using wavelet decomposed ensemble model,'' *Energy*, vol. 46, no. 1, pp. 564–574, Oct. 2012.

[42] B. Jin and X. Xu, ''Price forecasting through neural networks for crude oil, heating oil, and natural gas,'' *Meas., Energy*, vol. 1, Mar. 2024, Art. no. 100001.

[43] M. Hasan, M. Z. Abedin, P. Hajek, K. Coussement, M. N. Sultan, and B. Lucey, ''A blending ensemble learning model for crude oil price forecasting,'' *Ann. Oper. Res.*, vol. 2024, pp. 1–31, Jan. 2024.

[44] E. Kohlscheen, ''Forecasting oil prices with random forests,'' *Empirical Econ.*, vol. 66, no. 2, pp. 927–943, Feb. 2024.

[45] P. Moreno, I. Figuerola-Ferretti, and A. Muñoz, ''Forecasting oil prices with non-linear dynamic regression modeling,'' *Energies*, vol. 17, no. 9, p. 2182, May 2024.

[46] Y. Duan, Z. Ming, and X. Wang, ''A crude oil spot price forecasting method incorporating quadratic decomposition and residual forecasting,'' *J. Math.*, vol. 2024, no. 1, Apr. 2024, Art. no. 6652218.

[47] Z. Khan, T. Ekundayo, and S. A. Chaudhry, ''Comparative impact of days of the week and months of the year on WTI and Brent crude oil prices,'' *Pakistan J. Life Social Sci.*, vol. 22, no. 1, pp. 1–15, 2024.

[48] L. Liu, S. Zhou, Q. Jie, P. Du, Y. Xu, and J. Wang, ''A robust time-varying weight combined model for crude oil price forecasting,'' *Energy*, vol. 299, Jul. 2024, Art. no. 131352.

[49] A. Daghighiasli, A. Heidari, E. Abbasi, and M. Damankeshideh, ''Application of metasynthesis in identifying new combined genetic algorithm methods to solve problems in oil price forecasting,'' *Int. J. Knowl. Process. Stud.*, vol. 4, no. 3, pp. 38–51, 2024.

[50] J. Li, Z. Hong, C. Zhang, J. Wu, and C. Yu, ''A novel hybrid model for crude oil price forecasting based on MEEMD and mix-KELM,'' *Expert Syst. Appl.*, vol. 246, Jul. 2024, Art. no. 123104.

[51] K. A. Marill, ''Advanced statistics: Linear regression, Part I: Simple linear regression,'' *Academic Emergency Med.*, vol. 11, no. 1, pp. 87–93, Jan. 2004.

[52] I. Shah, H. Iftikhar, S. Ali, and D. Wang, ''Short-term electricity demand forecasting using components estimation technique,'' *Energies*, vol. 12, no. 13, p. 2532, Jul. 2019.

[53] R. E. Omay, ''The relationship between environment and income: Regression spline approach,'' *Int. J. Energy Econ. Policy*, vol. 3, no. 4, pp. 52–61, 2013.

[54] H. Iftikhar, J. E. Turpo-Chaparro, P. C. Rodrigues, and J. L. López-Gonzales, ''Forecasting day-ahead electricity prices for the Italian electricity market using a new decomposition–combination technique,'' *Energies*, vol. 16, no. 18, p. 6669, Sep. 2023.

[55] B. W. Silverman, ''Some aspects of the spline smoothing approach to non-parametric regression curve fitting,'' *J. Roy. Stat. Soc. Ser. B, Methodol.*, vol. 47, no. 1, pp. 1–21, Sep. 1985.

[56] H. Iftikhar, J. E. Turpo-Chaparro, P. C. Rodrigues, and J. L. López-Gonzales, ''Day-ahead electricity demand forecasting using a novel decomposition combination method,'' *Energies*, vol. 16, no. 18, p. 6675, Sep. 2023.

[57] W. S. Cleveland and S. J. Devlin, ''Locally weighted regression: An approach to regression analysis by local fitting,'' *J. Amer. Stat. Assoc.*, vol. 83, no. 403, p. 596, Sep. 1988.

[58] J. Fox, *Nonparametric Simple Regression: Smoothing Scatterplots*. Newbury Park, CA, USA: Sage, 2000.

[59] G. M. Jenkins and G. E. P. Box, *Time Series Analysis: Forecasting and Control*. Upper Saddle River, NJ, USA: Prentice-Hall, 1976.

[60] I. J. Myung, ''Tutorial on maximum likelihood estimation,'' *J. Math. Psychol.*, vol. 47, no. 1, pp. 90–100, Feb. 2003.

[61] O. Surakhi, M. A. Zaidan, P. L. Fung, N. Hossein Motlagh, S. Serhan, M. AlKhanafseh, R. M. Ghoniem, and T. Hussein, ''Time-lag selection for time-series forecasting using neural network and heuristic algorithm,'' *Electronics*, vol. 10, no. 20, p. 2518, Oct. 2021.

[62] P. Newbold, ''ARIMA model building and the time series analysis approach to forecasting,'' *J. Forecasting*, vol. 2, no. 1, pp. 23–35, Jan. 1983.

[63] R. H. Shumway, D. S. Stoffer, R. H. Shumway, and D. S. Stoffer, ''ARIMA models,'' in *Time Series Analysis and Its Applications: With R Examples*. Springer, 2017, pp. 75–163.

[64] C. Grillenzoni, ''ARIMA processes with ARIMA parameters,'' *J. Bus. Econ. Statist.*, vol. 11, no. 2, pp. 235–250, Apr. 1993.

[65] R. G. Brown, *Exponential Smoothing for Predicting Demand*. Boston, MA, USA: little, 1956.

[66] C. C. Holt, ''Forecasting seasonals and trends by exponentially weighted moving averages,'' *Int. J. Forecasting*, vol. 20, no. 1, pp. 5–10, Jan. 2004.

[67] E. S. Gardner, ''Exponential smoothing: The state of the art,'' *J. Forecasting*, vol. 4, no. 1, pp. 1–28, Jan. 1985.

[68] D. Effrosynidis, E. Spiliotis, G. Sylaios, and A. Arampatzis, ''Time series and regression methods for univariate environmental forecasting: An empirical evaluation,'' *Sci. Total Environ.*, vol. 875, Jun. 2023, Art. no. 162580.

[69] L. Wasserman, *All of Nonparametric Statistics*. Cham, Switzerland: Springer, 2006.

[70] M. Álvarez-Díaz, ''Is it possible to accurately forecast the evolution of Brent crude oil prices? An answer based on parametric and nonparametric forecasting methods,'' *Empirical Econ.*, vol. 59, no. 3, pp. 1285–1305, Sep. 2020.

[71] T. Taskaya-Temizel and M. C. Casey, ''A comparative study of autoregressive neural network hybrids,'' *Neural Netw.*, vol. 18, nos. 5–6, pp. 781–789, Jul. 2005.

[72] A. Tealab, H. Hefny, and A. Badr, ''Forecasting of nonlinear time series using ANN,'' *Future Comput. Informat. J.*, vol. 2, no. 1, pp. 39–47, Jun. 2017.

[73] L. Breiman, ''Random forests,'' *Mach. Learn.*, vol. 45, pp. 5–32, Oct. 2001.

[74] S. Goli, H. Mahjub, J. Faradmal, and A.-R. Soltanian, ''Performance evaluation of support vector regression models for survival analysis: A simulation study,'' *Int. J. Adv. Comput. Sci. Appl.*, vol. 7, no. 6, pp. 1–15, 2016.

[75] J. D. Borrero and J. Mariscal, ''Elevating univariate time series forecasting: Innovative SVR-empowered nonlinear autoregressive neural networks,'' *Algorithms*, vol. 16, no. 9, p. 423, Sep. 2023.

[76] A. Patle and D. S. Chouhan, ''SVM kernel functions for classification,'' in *Proc. Int. Conf. Adv. Technol. Eng. (ICATE)*, Jan. 2013, pp. 1–9.

[77] D.-X. Zhou and K. Jetter, ''Approximation with polynomial kernels and SVM classifiers,'' *Adv. Comput. Math.*, vol. 25, nos. 1–3, pp. 323–344, Jul. 2006.

[78] J.-L. Zhang, Y.-J. Zhang, and L. Zhang, ''A novel hybrid method for crude oil price forecasting,'' *Energy Econ.*, vol. 49, pp. 649–659, May 2015.

[79] G. A. Busari and D. H. Lim, ''Crude oil price prediction: A comparison between AdaBoost-LSTM and AdaBoost-GRU for improving forecasting performance,'' *Comput. Chem. Eng.*, vol. 155, Dec. 2021, Art. no. 107513.

[80] A. H. Vo, T. Nguyen, and T. Le, ''Brent oil price prediction using bi-LSTM network,'' *Intell. Autom. Soft Comput.*, vol. 26, no. 4, pp. 1307–1317, 2020.

[81] D. T. Phan, ''Comparison performance of spectrogram and scalogram as input of acoustic recognition task,'' in *Proc. Future Inf. Commun. Conf.*, in Lecture Notes in Networks and Systems, Cham, Switzerland, K. Arai, Ed. Springer, Jan. 2025, pp. 660–673.

[82] D. T. Phan, ''Reduce computational complexity for continuous wavelet transform in acoustic recognition using hop size,'' in *Proc. Int. Symp. Electron. Telecommun. (ISETC)*, Nov. 2024, pp. 1–4.

HASNAIN IFTIKHAR received the B.Sc. and M.Sc. degrees in statistics from the University of Peshawar, Pakistan, and the M.Phil. degree from Quaid-i-Azam University, Islamabad. He is currently pursuing the Ph.D. degree in statistics with the University of Peshawar. He is also a Subject Specialist of statistics with the Education Department, Peshawar, and a Young Advisor at the Facultad de Ingeniería y Arquitectura, Universidad Peruana Unión, Peru. He reviewed articles for several international journals, including IEEE Access, Elsevier, Springer, *PeerJ*, Sage, Royal Society, World Scientific, and Universal Publisher. His research interests include time series analysis, regression analysis, energy economics, biostatistics, and applied statistics.

MOIZ QURESHI received the M.Phil. degree in statistics from Quaid-e-Azam University, Islamabad. He is currently pursuing the Ph.D. degree in statistics with the University of Sindh, Jamshoro, Pakistan. He is a Lecturer of statistics with the Government Boys Degree College Tandojam, Hyderabad, Sindh, Pakistan, and a Visiting Lecturer with ISRA University, Hyderabad, Pakistan. He has reviewed several national and international journals, including Elsevier, Springer, *PeerJ*, and Sage Publisher. His research interests include statistical process control, time series analysis, machine learning, artificial intelligence, regression analysis, biostatistics and epidemiology, business statistics, financial analysis, and econometrics. He has also published various research articles in well-reputed international journals.

PAULO CANAS RODRIGUES received the Ph.D. degree in statistics from the Nova University of Lisbon, Portugal, in 2012, and the Aggregation (Habilitation) degree in mathematics, with a specialization in statistics and stochastic processes from Lisbon University, Portugal, in 2019. He is currently a Professor of statistics with the Federal University of Bahia and the Head and the Principal Investigator of the Statistical Learning Laboratory (SaLLy). His research interests include statistical learning, time series forecasting, and data science in general.

MUHAMMAD USMAN IFTIKHAR is currently pursuing the master's degree in data science with Deakin University, Australia. His work explores machine learning use cases with a focus on time series forecasting, predictive analytics, and computer vision applications.

JAVIER LINKOLK LÓPEZ-GONZALES (Member, IEEE) received the B.Sc. degree in statistical and informatics engineering from Universidad Peruana Unión (UPeU), Peru, the M.Sc. degree in metrology from the Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil, and the Ph.D. degree in statistics from the Universidad de Valparaíso (UV), Chile. He is currently a Full Professor with the Universidad Peruana Unión. In addition, he is qualified as a RENACYT Researcher. His research interests include pattern recognition in machine learning, air pollution with deep learning techniques, and time series with singular spectrum analysis.

HASNAIN IFTIKHAR is currently pursuing the master's degree in data science with Deakin University, Australia. His research interests include machine learning applications, particularly time series forecasting, predictive modeling, and algorithm development for data-driven decision-making.